

EXHIBIT 27

A solid black horizontal bar used for redaction, positioned below the exhibit title.

Exhibit 27C: U.S. Patent No. 12,412,924 – LGES

This claim chart shows how each element of the independent of U.S. Patent No. 12,412,924 (“the ’924 Patent”) is met by the LG Energy Solution (“LGES”) [REDACTED], either literally or under the doctrine of equivalents. The materials referenced, and the citations, are merely exemplary and are not exhaustive. LGES reserves the right to rely on different or additional sources and/or different or additional citations to sources that further illustrate a domestic industry. The evidence of domestic industry provided below is representative in nature and subject to change as discovery commences, including expert discovery. Further, the ALJ has not yet construed certain claim terms of the ’924 patent. Accordingly, LGES also expressly reserves the right to supplement or amend its claim charts as discovery progresses.

Claims	Exemplary Evidence of Infringement
1[pre]. A separator for a lithium battery comprising:	The [REDACTED] includes a separator for a lithium battery.  As indicated by LGES, the [REDACTED] is an NCM battery, which is a lithium-ion battery. Materials NCM(A) > https://www.lgensol.com/en/business/factor-cylindrical-46 (last accessed June 18, 2026).

NCM Battery

NCM batteries are lithium-ion batteries that use lithium nickel-cobalt-manganese oxide (LiNiCoMnO_2) as the cathode material. By combining nickel (Ni), cobalt (Co), and manganese (Mn) in the cathode, they achieve an optimized balance of performance and stability.

<https://www.lgensol.com/en/business/subject-ncm> (last accessed June 18, 2026).

As shown below, the [REDACTED] includes a separator.

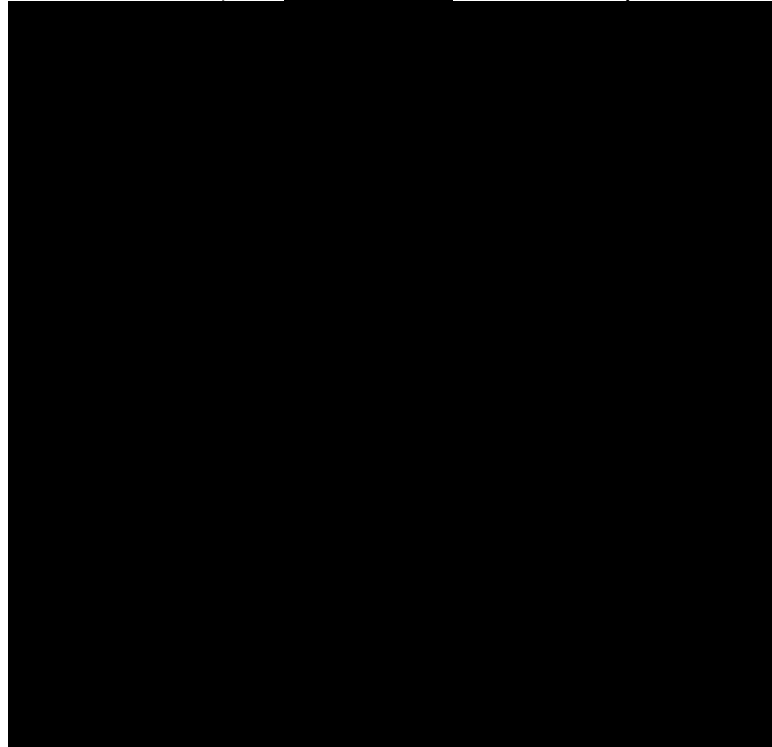
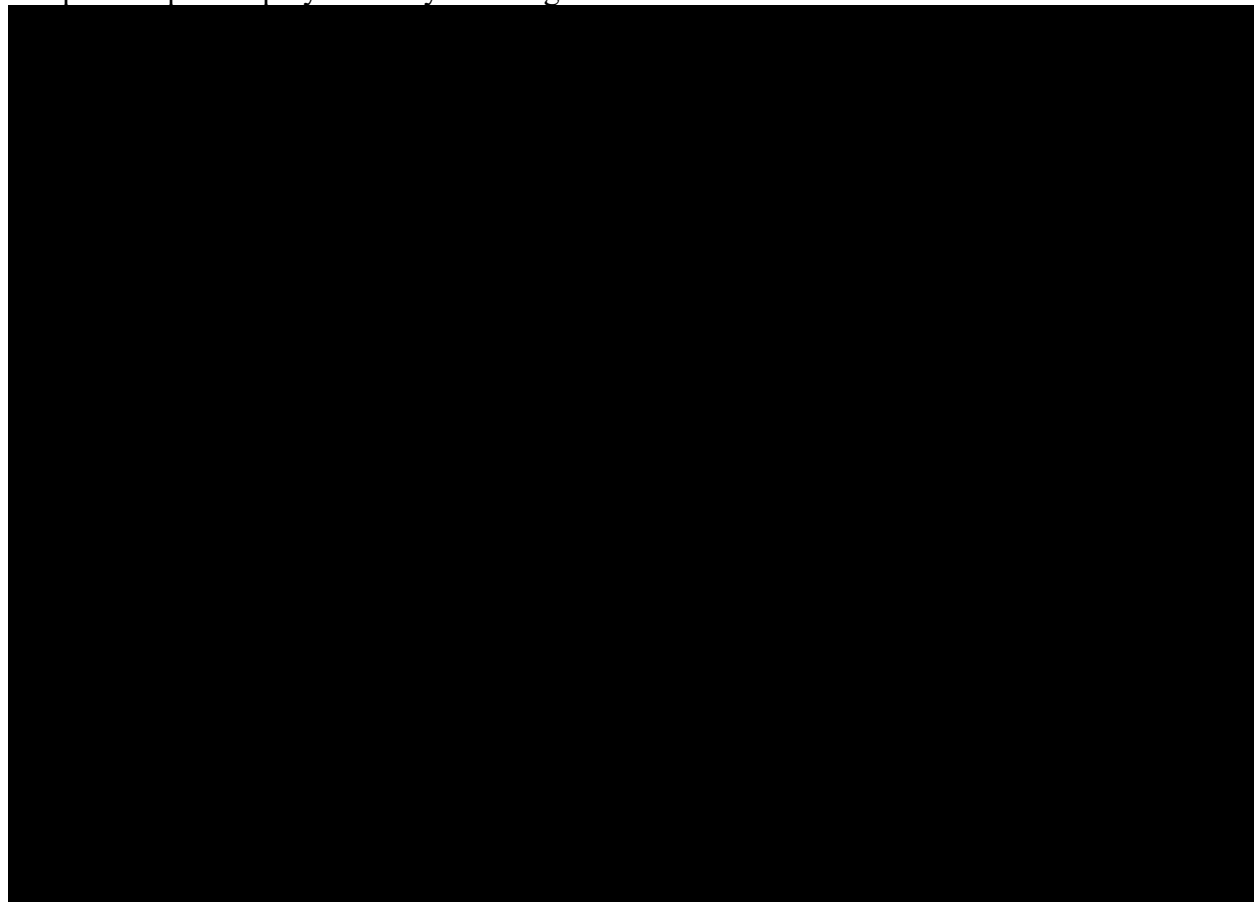


Image of the disassembled electrolyte assembly from the [REDACTED] (annotations added).

1[a]. a porous polymeric layer having a first side and a second side;

The [REDACTED]'s separator includes a porous polymeric layer having a first side and a second side.

As shown in the cross-section scanning electron microscopy (“SEM”) image below, the separator comprises a porous polymeric layer having a first side and a second side.

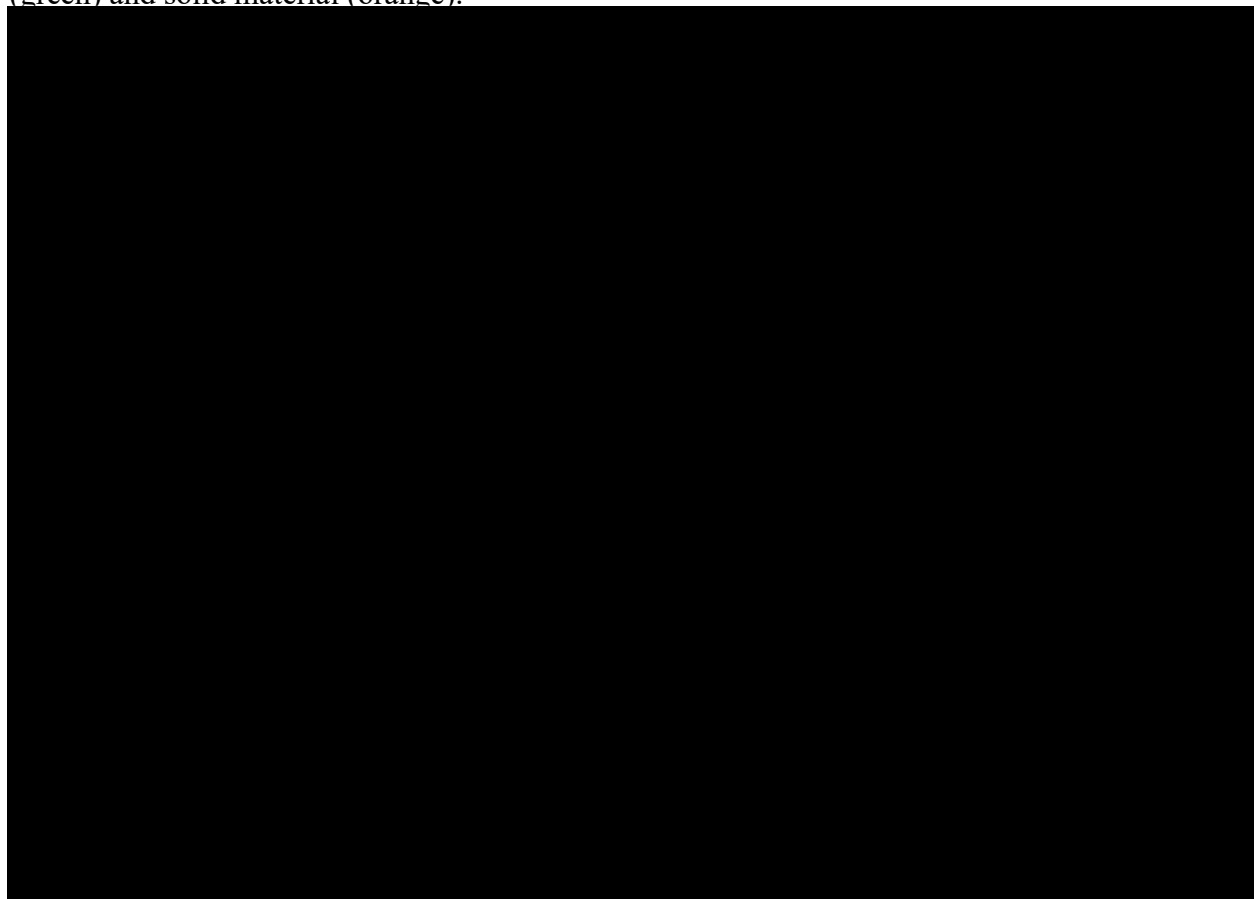


SEM image of the cross section of [REDACTED] milled by a focused ion beam (“FIB”) (annotations added).

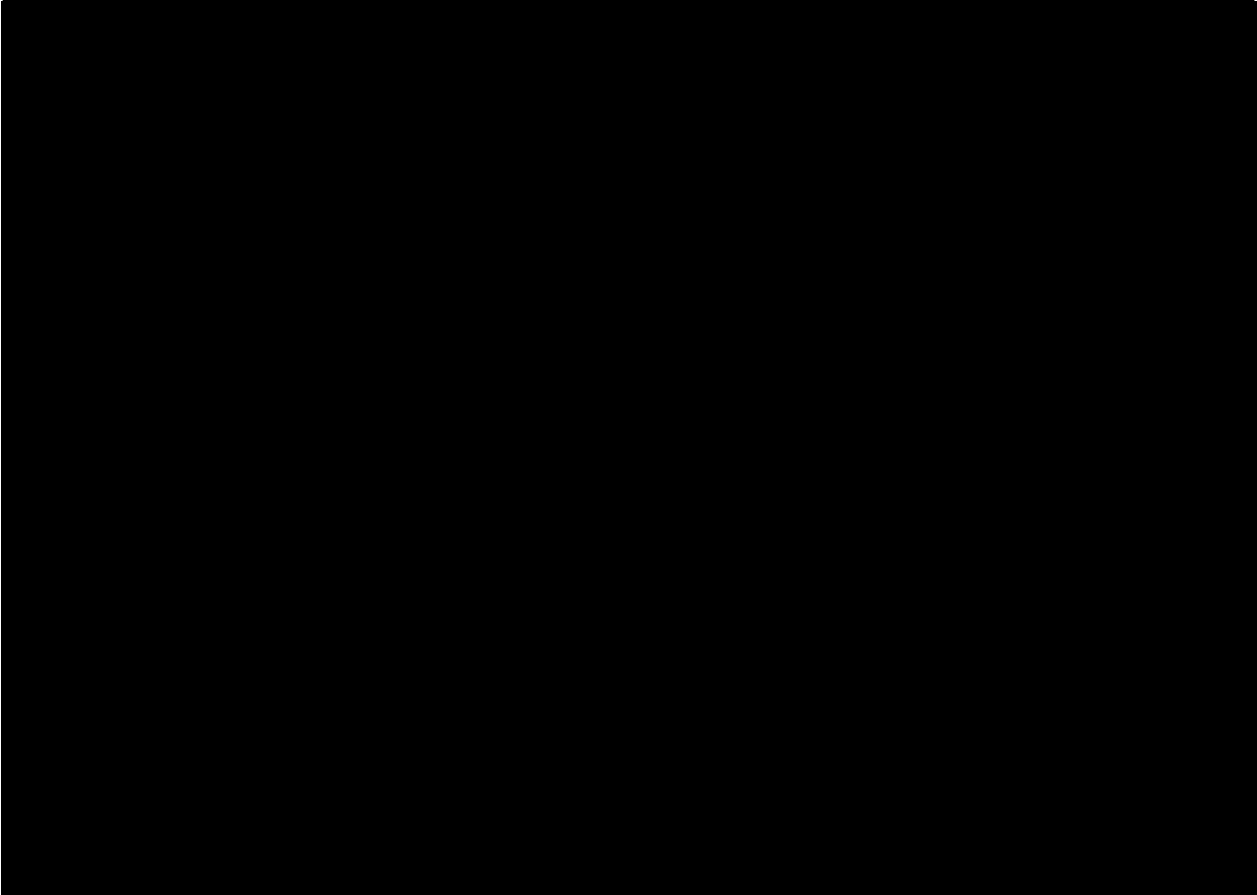
1[b]. a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material, wherein

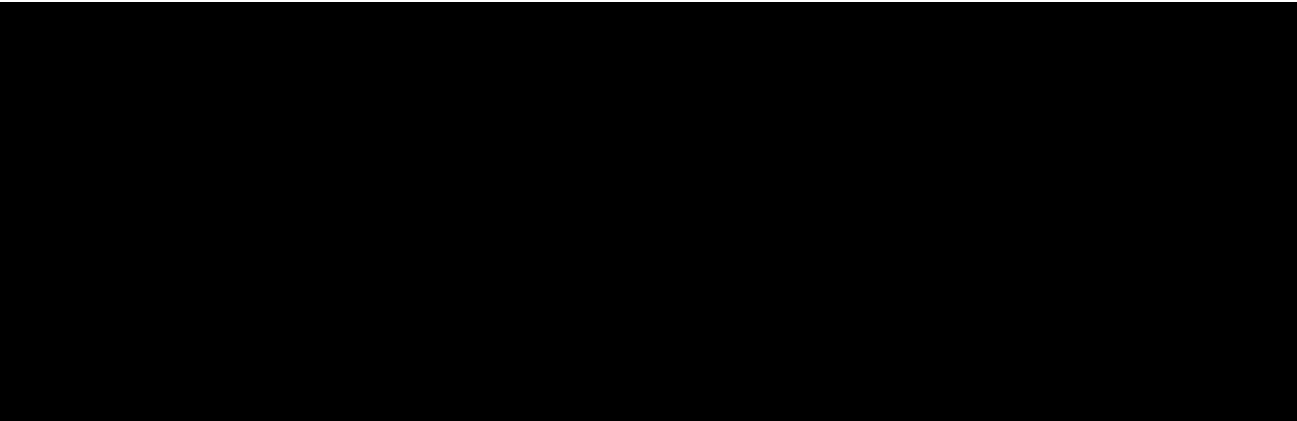
The [REDACTED]'s separator includes a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material.

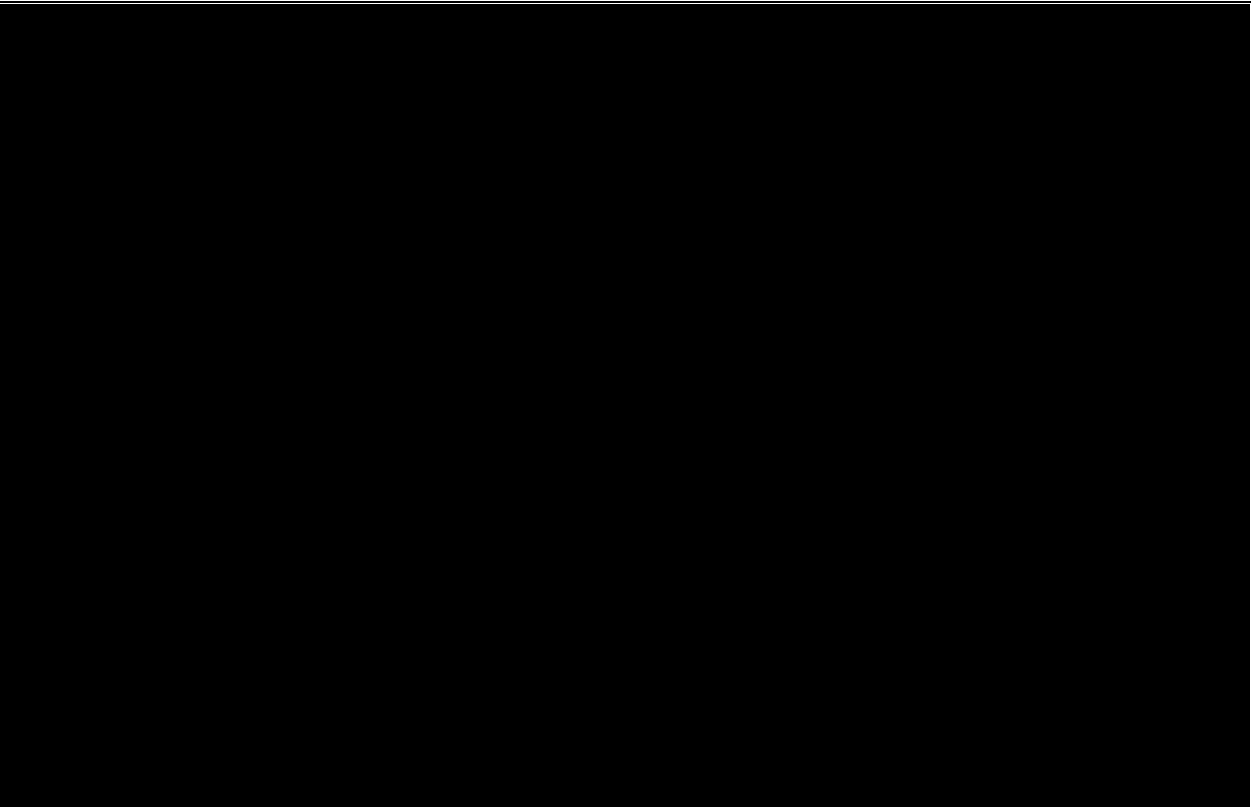
As shown in the cross-section SEM image below, the separator shows a first layer adjacent to the first side of the porous polymeric layer. The image further shows that the first layer includes pores (green) and solid material (orange).



SEM image of the FIB cross-section of [REDACTED] (annotations added).

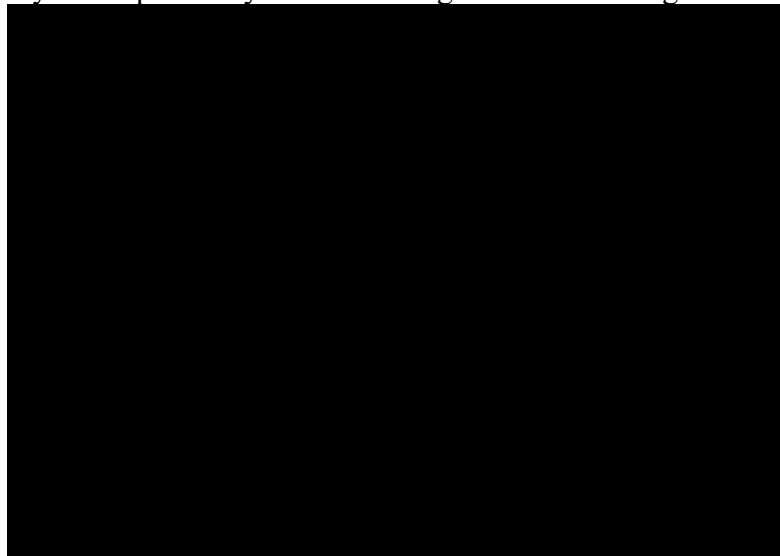
	When viewed from top down, the first layer under SEM also shows pores and solid material.  SEM image of the first layer of the [REDACTED] (annotations added).
1[c]. (a) the solid material comprises inorganic oxide particles and at least one polymer;	The [REDACTED]'s separator includes solid material that comprises inorganic oxide particles and at least one polymer.

	Scanning electron microscopy combined with energy dispersive spectroscopy (SEM-EDS) analysis of the solid material confirms the presence of an aluminum oxide compound () with strong aluminum and oxygen peaks.  SEM-EDS analysis of the solid material in the first layer of the separator in the . X-ray diffraction (XRD) analysis further confirms the solid material comprises inorganic oxide particles ().
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	 XRD analysis of the solid material in the first layer of the separator in the [REDACTED]. Further, the [REDACTED] particles adhere to the porous polymeric layer of the separator, as shown in the cross-section SEM images above. On information and belief, the first layer includes at least one polymer that functions as a binder to allow the corundum particles to adhere to the surface of the porous polymeric layer.
1[d]. (b) a plurality of the inorganic oxide particles in the first layer comprise	The [REDACTED]'s separator includes a plurality of the inorganic oxide particles in the first layer that comprise crystallites having a size in the range of 5 nm to 90 nm.

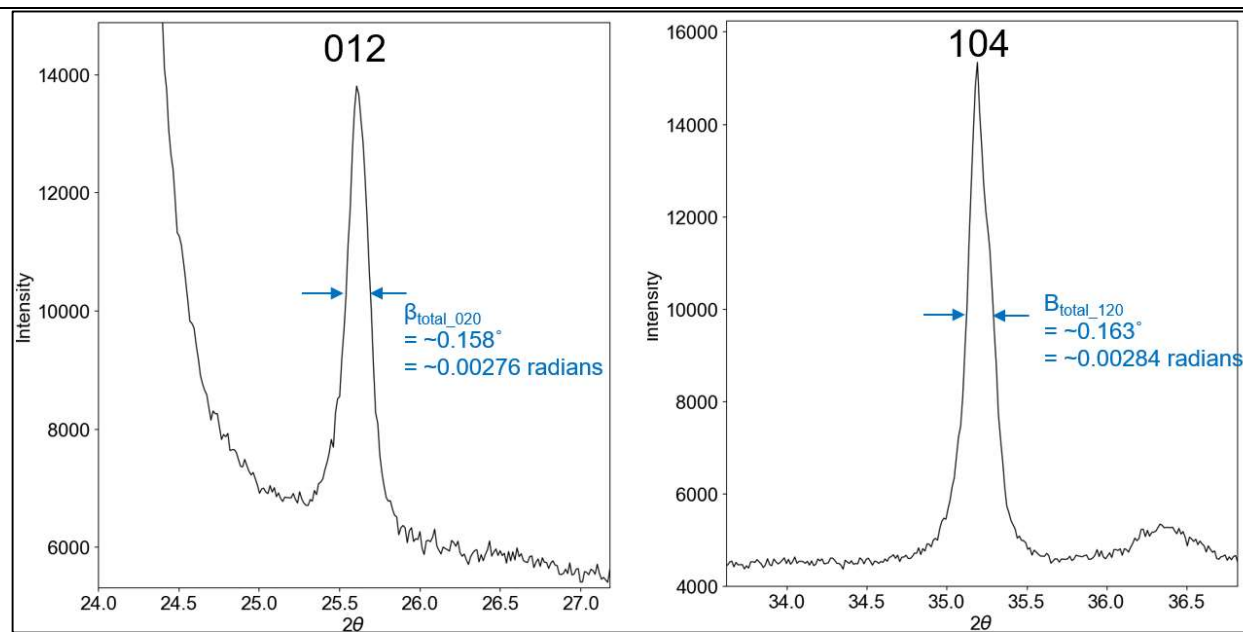
crystallites having a size in the range of 5 nm to 90 nm;

As shown in the top-down SEM images below, a plurality of the inorganic oxide particles in the first layer comprises crystallites having a size in the range of 5 nm to 90 nm, *e.g.*, [REDACTED] nm.



SEM image of the first layer of the [REDACTED] (annotation added).

Further, using the Scherrer equation, the average crystallite size of the corundum particles in the first layer is estimated to be [REDACTED] nm.



XRD analysis for corundum in the first layer of the [REDACTED] separator.

Scherrer Equation:

$$D = \frac{K\lambda}{\beta_{sample} \cos(\theta)}$$

D = mean crystallite size

K = shape factor, assumed to be 0.9

λ = wavelength of source, 0.154 nm

β_{sample} = corrected full width at half maximum*

θ = Bragg angle

$$\beta_{total}^{1.5} = \beta_{sample}^{1.5} + \beta_{inst}^{1.5}$$

$\beta_{inst} = 0.00161$ rad (same value for both peaks)

$\beta_{sample\ 012} =$ [REDACTED] rad (using [REDACTED] peak)

$\beta_{sample\ 104} =$ [REDACTED] rad (using [REDACTED] peak)

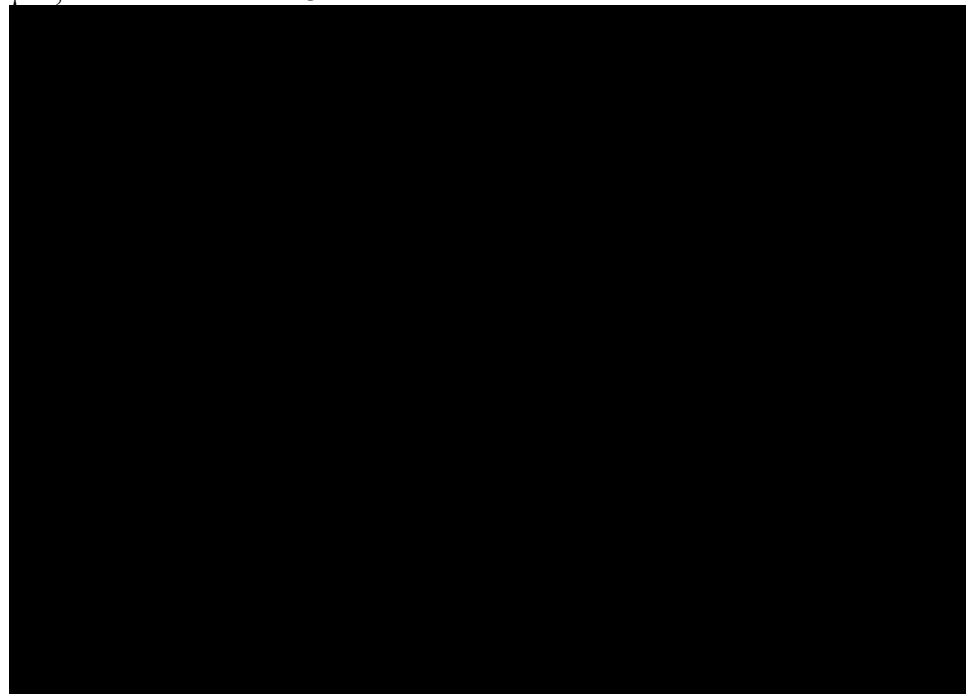
$$D = \text{[REDACTED]} \text{ (using [REDACTED] peak at } 2\theta = \text{[REDACTED]}^\circ)$$
$$D = \text{[REDACTED]} \text{ (using [REDACTED] peak at } 2\theta = \text{[REDACTED]}^\circ)$$

Accordingly, the mean crystallite size of the corundum was calculated to be [REDACTED] nm using the XRD results for the [REDACTED] and [REDACTED] peaks and the Scherrer Equation after correcting for the instrument broadening.

1[e]. (c) a thickness of the first layer is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm; and

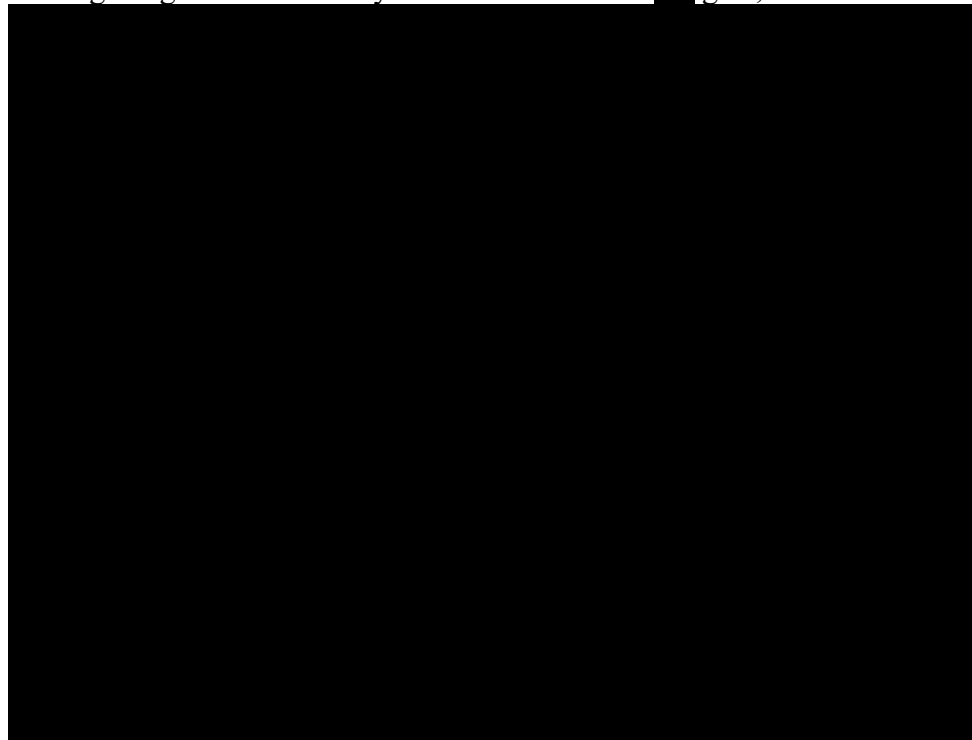
The [REDACTED]'s separator has a thickness of the first layer that is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm.

As shown in the cross-section SEM image below, a thickness of the first layer is estimated to be [REDACTED] μm , which is less than 3 microns.



SEM image of the FIB cross-section of [REDACTED] (annotations added).

Based on thermogravimetric analysis (TGA) of the coated separator in the [REDACTED], a total coating weight of the first layer is estimated to be [REDACTED] gsm, which is less than 6.0 gsm.



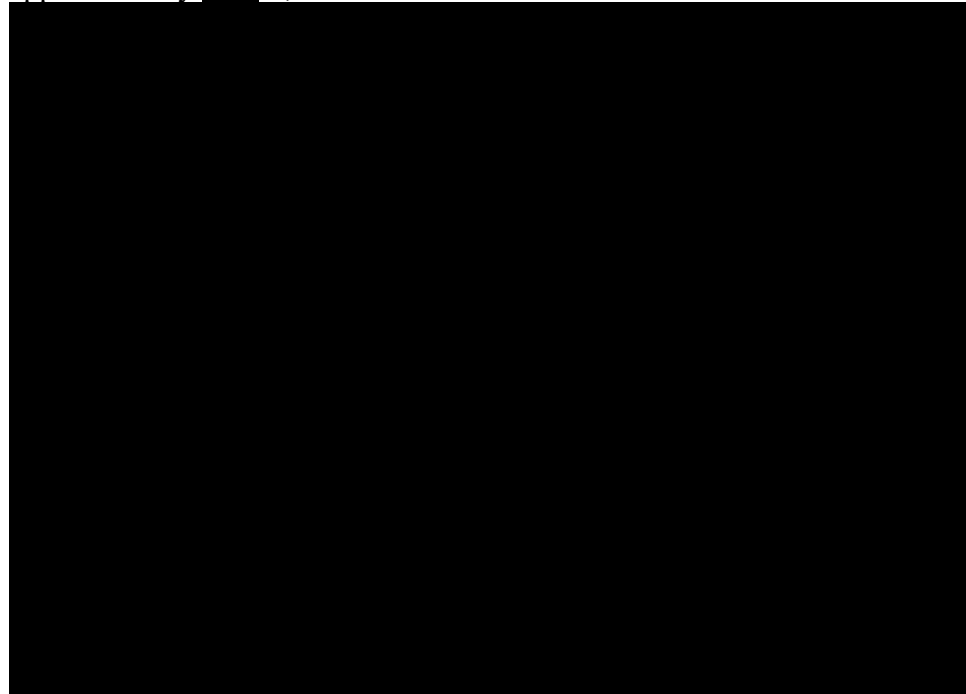
TGA analysis of the separator of the [REDACTED]

Specifically, the average mass of the 8 mm diameter separator samples was [REDACTED] mg, which corresponds to a total separator footprint mass of [REDACTED] g/m². The relative mass ratio of polymeric-to-inorganic material was determined to be [REDACTED] based on TGA. Therefore, the coating mass was calculated as [REDACTED] g/m², which is less than 6.0 g/m².

1[f]. (d) a ratio of the thickness of the first layer to a total thickness of the separator is 10% to 80%.

The [REDACTED]'s separator has a ratio of the thickness of the first layer to a total thickness of the separator between 10% and 80%.

As shown below, a ratio of the thickness of the first layer to the total thickness of the separator is approximately [REDACTED]%, which is between 10% and 80%.



SEM image of the FIB cross-section of [REDACTED] (annotations added).